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1 LaTeX-ready snippet (minimal, axiomatic)

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I have reworked the book from Gary Harper to have a full LaTeX version with correct BibTex references. For an lecture “Geometric Algebra(GA)” I need an clear axiomatic formulation of all axioms and principles defined in GA without any historic comments or comparison to older variants. I need this for students that have never seen vector calculus and do not need any hints to old stuff. Thid is only the first step, the next step is to provide a discrete differential approach in the GA setup (based on work Keenan Crane).

1.2 ChatGPT

I’ll extract the GA foundations from the LaTeX/PDF you provided and then give you a clean, “axioms → definitions → derived laws” formulation suitable as the opening of your lecture notes, with no historical detours.

Thought for 1m 51s

Below is a clean axioms $\rightarrow$ definitions $\rightarrow$ derived laws formulation you can drop into your lecture notes. It assumes only basic linear algebra (vector spaces, bilinear maps). No historical remarks, no “older variants”.

Axiomatic core of Geometric Algebra 0. Primitives (given data)

A field of scalars: $\mathbb{R}$.

A finite-dimensional real vector space V .

A symmetric bilinear form (the “metric”)

$$g : V \times V \rightarrow \mathbb{R}, \quad g(u, v) = g(v, u),$$

not assumed positive definite (signature is allowed).

1. Axioms (defining the geometric algebra) Axiom GA1 (Algebra structure)

There exists a real associative unital algebra G with product $(A, B) \mapsto AB$ and unit $1 \in G$. So for all $A, B, C \in G$ and $\alpha, \beta \in \mathbb{R}$:

Associativity: $(AB)C = A(BC)$.

Distributivity: $A(B + C) = AB + AC$, $(A + B)C = AC + BC$.

Scalar bilinearity: $(\alpha A)(\beta B) = (\alpha\beta)(AB)$.

Unit: $1A = A1 = A$.

Axiom GA2 (Vector embedding and generation)

There is an injective linear map $\iota : V \hookrightarrow G$. We identify $v \in V$ with $\iota(v) \in G$. The algebra G is generated by $\mathbb{R} \cup V$ under $+$ and the product.

Axiom GA3 (Clifford / quadratic relation)

For all $u, v \in V$,

$$uv + vu = 2g(u, v)1.$$

Equivalently, for all $v \in V$,

$$v^2 = g(v, v)1.$$

Definition (Geometric algebra): The pair (G, V) satisfying GA1-GA3 is the geometric algebra of (V, g) , also denoted $G = \text{Cl}(V, g)$.

2. Immediate consequences (“principles” you can use as rules) C1 (Orthogonal vectors anticommute)

If $g(u, v) = 0$, then $uv = -vu$.

C2 (Geometric product splits into symmetric + antisymmetric parts for vectors)

For $u, v \in V$, define:

$$u \cdot v := g(u, v) \in \mathbb{R}, \quad u \wedge v := \frac{1}{2}(uv - vu).$$

Then

$$uv = u \cdot v + u \wedge v.$$

$u \cdot v$ is a scalar (grade 0).

$u \wedge v$ is a bivector (grade 2), and $u \wedge v = -v \wedge u$.

C3 (Invertibility of non-null vectors)

If $v \in V$ with $g(v, v) \neq 0$, then v is invertible in G and

$$v^{-1} = \frac{v}{g(v, v)}.$$

3. Grades and multivectors (structure you'll need for everything later) D1 (Exterior powers inside G)

Define $\Lambda^0 V := \mathbb{R}$, $\Lambda^1 V := V$. For $k \geq 2$, define $\Lambda^k V$ as the span of all wedge products

$$v_1 \wedge \cdots \wedge v_k$$

formed using the wedge from C2 (extended by multilinearity and associativity).

D2 (Grade decomposition)

As a vector space,

$$G \cong \bigoplus_{k=0}^n \Lambda^k V, \quad n = \dim V.$$

An element $A \in G$ is a multivector, and decomposes uniquely:

$$A = \sum_{k=0}^n \langle A \rangle_k, \quad \langle A \rangle_k \in \Lambda^k V.$$

$\langle A \rangle_k$ is the grade- k part (grade projection).

Principle P1 (Grade-selection product rule for homogeneous elements)

If $A_r \in \Lambda^r V$ and $B_s \in \Lambda^s V$, then the geometric product $A_r B_s$ contains only grades $|r - s|, |r - s| + 2, \dots, r + s$.

This is the “selection rule” that lets you define all other products as grade parts.

4. Derived operations (defined, not axiomatized)

These are standard, but you should pick one convention and stick to it.

D3 (Outer product for general homogeneous elements)

For $A_r \in \Lambda^r V$, $B_s \in \Lambda^s V$,

$$A_r \wedge B_s := \langle A_r B_s \rangle_{r+s}.$$

Extended by bilinearity to all multivectors.

D4 (Left contraction / inner product for homogeneous elements)

For $A_r \in \Lambda^r V$, $B_s \in \Lambda^s V$ with $r \leq s$,

$$A_r \lrcorner B_s := \langle A_r B_s \rangle_{s-r}.$$

If $r > s$, set $A_r \lrcorner B_s := 0$. Extend bilinearly.

(For vectors u, v , this recovers $u \lrcorner v = u \cdot v = g(u, v)$.)

D5 (Three canonical involutions)

For homogeneous $A_r \in \Lambda^r V$:

Grade involution: $\widehat{A_r} = (-1)^r A_r$.

Reversion: $\widetilde{A}_r = (-1)^{r(r-1)/2} A_r$.

Clifford conjugation: $\overline{A} := \widetilde{A}$.

Extend linearly to general multivectors $A = \sum \langle A \rangle_r$.

5. Metric-dependent duality (needed later for Hodge star / DEC)

Assume g is nondegenerate and V is oriented.

D6 (Pseudoscalar and dual)

Let $I \in \Lambda^n V$ be a chosen nonzero unit pseudoscalar (orientation element). If I is invertible (true for nondegenerate g), define the dual:

$$A^* := A I^{-1}.$$

This is the GA version of the Hodge dual; it is metric- and orientation-dependent.

6. Versors and orthogonal transformations (GA's "action principle") D7 (Versor)

A versor is a product of invertible vectors:

$$V = v_1 v_2 \cdots v_k, \quad g(v_i, v_i) \neq 0.$$

It is invertible, with inverse V^{-1} existing in G .

Principle P2 (Sandwich action)

Define an action on vectors (and then on multivectors) by

$$x \mapsto x' := V x V^{-1}.$$

This preserves the quadratic form g (orthogonal transformation).

Special case: reflections and rotations

Reflection in a hyperplane with unit normal n ($n^2 = \pm 1$):

$$x' = -n x n^{-1}.$$

Rotor R : an even versor with $R \widetilde{R} = 1$. Then

$$x' = R x \widetilde{R}$$

is a proper orthogonal transformation.

LaTeX-ready snippet (minimal, axiomatic)

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\section{Axioms of Geometric Algebra}

Let  $V$  be a finite-dimensional real vector space and let
 $g: V \times V \rightarrow \mathbb{R}$  be a symmetric bilinear form.

\begin{axiom}[GA1: associative unital algebra]
There exists a real associative unital algebra  $G$  with product  $(A, B) \mapsto AB$ 
and unit  $1 \in G$ .
\end{axiom}

\begin{axiom}[GA2: vector embedding and generation]
There exists an injective linear map  $\iota: V \hookrightarrow G$ .
We identify  $v \in V$  with  $\iota(v) \in G$  and require that  $G$  is generated
by  $\mathbb{R} \cup V$  under addition and multiplication.
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15 \end{axiom}
16
17 \begin{axiom}[GA3: Clifford relation]
18 For all $u,v\in V$,
19 \[
20 uv+vu = 2\langle u,v\rangle, 1.
21 \]
22 \end{axiom}
23
24 \begin{definition}[Inner and outer product of vectors]
25 For $u,v\in V$ define
26 \[
27 u\cdot v := \langle u,v\rangle, \quad u\wedge v := \tfrac{1}{2}(uv-vu),
28 \]
29 so that $uv = u\cdot v + u\wedge v$.
30 \end{definition}
31
32
33 \begin{definition}[Grade decomposition]
34 Let $n=\dim V$. As a vector space,
35 $G\cong \bigoplus_{k=0}^n \Lambda^k V$.
36 Every $A\in G$ decomposes uniquely as
37 $A=\sum_{k=0}^n \langle A, e_k\rangle e_k$ with $\langle A, e_k\rangle\in \Lambda^k V$.
38 \end{definition}

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How this sets you up for the “discrete differential” step

For the Keenan-Crane-style discrete story (DEC), the objects you will need are exactly:

$\wedge$ (for oriented k -cells / discrete forms),

contraction (pairing, boundary/coboundary adjoints),

duality $A^* = AI^{-1}$ (Hodge star / primal-dual mesh coupling),

grade projections (to keep operators type-correct).

When you’re ready, I can map: k -cochains $\leftrightarrow k$ -vectors/forms, d as a discrete exterior derivative, $\star$ via pseudoscalar/metric, and the codifferential via adjoints—all expressed in GA notation and kept “students-first”.

Entities referenced: Gary Harper, Keenan Crane